

HATGAL, Mongolia

WMO: 442070

Lat: 50.4394N

Lon: 100.1508E

Elev: 5472

StdP: 12.01

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.8	-30.3	-40.2	0.7	-34.5	-35.9	0.9	-30.0	23.4	3.6	22.4	5.2	3.3	270	0.801

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	74.7	56.7	71.2	55.1	68.1	53.9	59.2	70.3	57.2	67.7	55.6	65.2	9.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.8	78.5	63.3	52.9	73.0	61.4	51.1	68.4	60.3	28.4	70.2	27.0	67.6	25.8	65.3	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.1	19.8	17.9	DB	-37.5	81.4	6.2	3.5	-41.9	83.9	-45.5	86.0	-48.9	88.0	-53.4	90.6	
			WB	-38.3	63.0	5.9	2.3	-42.6	64.7	-46.0	66.0	-49.3	67.3	-53.6	69.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	-9.1	-2.2	13.1	30.2	40.7	51.1	56.0	53.0	42.4	27.9	10.1	-3.0
(7)		DBStd	24.22	10.13	11.00	11.49	9.14	8.10	5.87	4.58	5.35	7.06	9.53	11.56	11.23
(8)		HDD50	9193	1831	1462	1143	595	302	56	6	30	241	685	1198	1644
(9)		HDD65	14239	2296	1882	1608	1044	754	416	282	372	678	1150	1648	2109
(10)		CDD50	433	0	0	0	1	14	90	191	124	13	0	0	0
(11)		CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
(12)		CDH74	124	0	0	0	0	6	32	54	30	2	0	0	0
(13)		CDH80	12	0	0	0	0	0	1	6	5	0	0	0	0
(14)	Wind	WSAvg	7.0	6.1	6.5	7.9	8.4	8.1	6.8	6.0	6.9	7.4	7.7	6.8	
(15)	Precipitation	PrecAvg	11.3	0.0	0.0	0.1	0.4	0.6	1.9	3.3	3.1	1.3	0.5	0.2	0.1
(16)		PrecMax	17.0	0.4	0.2	0.6	2.4	2.5	5.9	7.8	7.7	4.8	2.5	0.8	0.4
(17)		PrecMin	7.8	0.0	0.0	0.0	0.0	0.0	0.2	0.7	0.7	0.1	0.0	0.0	0.0
(18)		PrecStd	2.6	0.1	0.0	0.1	0.4	0.5	1.0	1.7	1.5	0.8	0.5	0.2	0.1
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.7	30.1	49.3	63.0	74.4	78.6	80.2	77.4	70.7	57.5	42.2	25.8
(20)		MCWB	18.3	24.1	37.4	45.0	52.1	56.5	60.1	59.0	51.6	42.7	33.2	21.1	
(21)		2%	DB	15.4	24.9	42.2	56.1	68.4	73.0	74.7	72.1	65.4	52.2	36.0	20.8
(22)		MCWB	12.6	20.1	32.5	40.4	48.8	54.5	58.4	56.7	49.6	40.0	29.5	17.3	
(23)		5%	DB	10.8	20.2	37.1	51.3	63.5	69.1	71.0	68.4	61.0	47.4	31.3	17.0
(24)		MCWB	8.5	16.4	29.0	37.8	46.6	53.0	56.7	55.2	47.9	37.2	26.2	14.0	
(25)		10%	DB	6.6	15.1	32.1	47.0	58.6	65.5	67.5	65.1	56.7	43.3	27.0	13.2
(26)		MCWB	4.8	12.1	25.9	35.5	43.9	51.7	55.7	54.1	45.6	34.8	22.8	10.9	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	24.4	38.9	45.9	53.9	58.4	63.0	60.9	54.1	44.0	34.0	21.4
(28)		MDCB	21.8	29.9	49.1	61.1	71.8	74.0	76.1	73.6	65.9	56.5	41.3	25.2	
(29)		2%	WB	13.0	20.5	32.7	41.4	50.2	56.0	60.0	58.3	51.2	40.8	29.6	17.4
(30)		MDCB	15.8	25.2	41.6	54.7	66.7	69.2	71.0	69.3	62.8	51.5	35.8	20.8	
(31)		5%	WB	8.6	16.8	29.5	38.4	47.1	54.4	58.2	56.5	48.8	38.1	26.4	14.3
(32)		MDCB	10.7	20.6	36.9	50.3	61.8	66.4	68.2	66.1	58.7	47.6	31.3	17.1	
(33)		10%	WB	4.8	12.6	26.2	35.8	44.4	52.6	56.4	54.8	46.5	35.5	23.0	11.1
(34)		MDCB	6.6	15.5	32.4	46.5	57.8	63.7	65.6	63.8	55.5	43.8	27.1	13.4	
(35)	Mean Daily Temperature Range	MDBR	21.2	27.0	27.9	24.2	25.5	23.1	20.0	20.8	24.0	23.7	21.4	18.9	
(36)		5% DB	MCDBR	23.1	29.5	30.0	31.5	35.5	32.0	28.4	28.0	31.9	29.0	23.9	19.9
(37)		MCWBR	21.0	25.3	22.0	19.3	20.2	17.0	14.6	14.9	18.1	18.7	18.3	17.3	
(38)		5% WB	MCDBR	23.3	29.1	28.9	29.8	32.7	26.7	23.7	23.8	27.8	28.2	23.1	19.7
(39)		MCWBR	21.0	25.1	21.5	19.0	19.6	14.9	13.2	13.4	16.5	18.7	18.2	17.2	
(40)	Clear-Sky Solar Irradiance	taub	0.222	0.245	0.287	0.329	0.354	0.349	0.341	0.322	0.286	0.267	0.244	0.225	
(41)		taud	2.330	2.298	2.255	2.297	2.269	2.395	2.458	2.486	2.530	2.485	2.365	2.296	
(42)		Ebn at Noon	270	289	292	290	285	288	288	290	292	279	257	248	
(43)		Edh at Noon	20	27	34	37	40	36	33	31	26	22	19	17	
(44)	All-Sky Solar Radiation	RadAvg	475	806	1282	1721	1957	1964	1866	1612	1324	885	516	368	
(45)		RadStd	19	28	55	76	113	115	98	83	76	49	28	16	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-326	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page